



Proposal for an ultra-broadband polarization beam splitter using an anisotropy-engineered Mach-Zehnder interferometer on the x-cut lithium-niobate-on-insulator

HONGNAN XU, DAOXIN DAI, LIU LIU, AND YAOCHENG SHI*

Centre for Optical and Electromagnetic Research, State Key Laboratory for Modern Optical Instrumentation, Zhejiang University, Zijingang Campus, Hangzhou 310058, China

*yaocheng@zju.edu.cn

Abstract: We propose and theoretically demonstrate an integrated polarization beam splitter on the x-cut lithium-niobate-on-insulator (LNOI) platform. The device is based on a Mach-Zehnder interferometer with an anisotropy-engineered multi-section phase shifter. The phase shift can be simultaneously controlled for the TE and TM polarizations by engineering the length and direction of the anisotropic LNOI waveguide. For TE polarization, the phase shift is $-\pi/2$, while for TM polarization, the phase shift is $\pi/2$. Thus, the incident TE and TM modes can be coupled into different output ports. The simulation results show an ultra-high polarization extinction ratio of ~ 47.7 dB, a low excess loss of ~ 0.9 dB and an ultra-broad working bandwidth of ~ 200 nm. To the best of our knowledge, the proposed structure is the first integrated polarization beam splitter on the x-cut LNOI platform.

© 2020 Optical Society of America under the terms of the [OSA Open Access Publishing Agreement](#)

1. Introduction

The lithium niobate (LiNbO_3 , LN) has been regarded as a high-performance optical material for its strong electro-optic response, large nonlinear coefficient, wide transparency window and long-term stability. The conventional LN waveguides are usually based on the titanium-diffusion or proton-exchange. These LN waveguides have a low index contrast (< 0.02) and a large mode size, leading to the bulky device size, high driving voltage and low nonlinear efficiency. Recently, the lithium-niobate-on-insulator (LNOI) has emerged as a promising approach to realize the high-contrast LN waveguide with sub-wavelength light confinement [1–5]. The LNOI is formed by a LN thin film bonded on a silicon-dioxide (SiO_2) substrate, and the waveguides can be built by directly dry etching the LN layer. Over the recent years, the LNOI has been widely employed in various applications, e.g., high-quality optical cavities [6–8], high-speed modulators [9–14], efficient wavelength converters [15,16] and compact spectrometers [17].

The polarization beam splitter (PBS) is one of the most important polarization-handling devices for splitting and combining the transverse-electric (TE) and transverse-magnetic (TM) polarizations, which has been widely applied in the polarization diversity circuit (PDC) [18], polarization-division multiplexing (PDM) [19] and quantum-photonics integrated circuit [20]. Various structures have been demonstrated to realize the high-performance PBS, e.g., asymmetric directional couplers (ADCs) [21–24], Bragg gratings [25], multi-mode interference (MMI) couplers [26,27] and sub-wavelength grating (SWG) metamaterials [28–30]. Most of these PBSs are designed based on the silicon-on-insulator (SOI) platform. However, there is a lack of high-performance PBS on the x-cut LNOI platform, which makes it difficult to manipulate the polarizations in the LNOI-based photonic circuits. In Ref. [31], a five-channel adiabatic coupler has been utilized to realize the PBS on the z-cut LNOI platform with an off-plane optical axis. However, the x-cut LNOI with an in-plane optical axis is commonly considered as a better choice especially for the switching and modulation applications since the electrodes can be simply placed

besides the x-cut LNOI waveguide [9–11]. It should be noted that the mode effective index for the x-cut LNOI waveguide will change with the propagation direction due to the in-plane optical axis [32], which makes it challenging to design the PBS on the x-cut LNOI platform. In this paper, we propose an ultra-broadband PBS based on the x-cut LNOI. The proposed structure is based on a Mach-Zehnder interferometer (MZI) with an anisotropy-engineered multi-section phase shifter. The phase shift can be simultaneously controlled for the TE and TM polarizations by engineering the length and direction of the LNOI waveguide, so the incident TE and TM polarizations can be separated with an ultra-high polarization extinction ratio and a low excess loss over an ultra-broad wavelength band > 200 nm. Such LNOI-based PBS can find its applications by integrating with other LNOI devices, e.g., optical switches or high-speed modulators, to realize the polarization-transparent photonic circuits or high-performance PDM transmitters. To the best of our knowledge, the proposed structure is the first integrated PBS on the x-cut LNOI platform, which expands the tool box of LNOI-based photonic devices.

2. Design and analysis

Figure 1(a) shows the schematic of the MZI-based PBS, which is composed of a 1×2 MMI, a multi-section phase shifter (PS) and a 2×2 MMI. Here, all the MMIs are designed to be polarization-insensitive. Thus, both the incident TE₀ and TM₀ modes can be evenly split into two beams by the 1×2 MMI. The split beams are then combined by the 2×2 MMI. The TE₀ and TM₀ can be coupled into the different output ports (O₁ and O₂) due to the polarization-dependent phase shift in the anisotropy-engineered PS. The PS is composed of two phase shifting sections with different length (dL₁ and dL₂) and directions [see Figs. 1(a) and 1(b)]. In PS section #1, the light propagates along z-direction, while in PS section #2, the light propagates along y-direction. The optical axis is along z-direction, so the effective indices are distinct for the LNOI waveguide in section #1 and #2. Figure 1(b) shows the effective index distribution for the anisotropy-engineered PS. One can find that the effective length difference dL_{eff} is dependent on both dL₁ and dL₂:

$$dL_{eff} = n_z dL_1 - n_y dL_2, \quad (1)$$

where n_y and n_z are the effective indices for the guided mode propagating along y- and z-directions. It should be noted that the effective indices are also different between the TE₀ and TM₀ modes due to the waveguide birefringence and material anisotropy. Thus, the TE and TM effective length differences can be written as:

$$\begin{bmatrix} dL_{eff,TE} \\ dL_{eff,TM} \end{bmatrix} = \begin{bmatrix} n_{z,TE0} & -n_{y,TE0} \\ n_{z,TM0} & -n_{y,TM0} \end{bmatrix} \begin{bmatrix} dL_1 \\ dL_2 \end{bmatrix}, \quad (2)$$

where $dL_{eff,TE}$ and $dL_{eff,TM}$ are the effective length differences for the TE and TM polarizations, $n_{y,TE0}$ and $n_{z,TE0}$ are the effective indices for the TE₀ mode propagating along y- and z-directions, $n_{y,TM0}$ and $n_{z,TM0}$ are the effective indices for the TM₀ mode propagating along y- and z-directions. Thus, $dL_{eff,TE}$ and $dL_{eff,TM}$ can be simultaneously controlled by properly choosing dL₁ and dL₂ according to the following equations:

$$dL_1 = \Delta_1 / \Delta_0, \quad (3)$$

$$dL_2 = \Delta_2 / \Delta_0, \quad (4)$$

$$\Delta_0 = n_{z,TE0} \cdot n_{y,TM0} - n_{z,TM0} \cdot n_{y,TE0}, \quad (5)$$

$$\Delta_1 = dL_{eff,TE} \cdot n_{y,TM0} - dL_{eff,TM} \cdot n_{y,TE0}, \quad (6)$$

$$\Delta_2 = dL_{eff,TE} \cdot n_{z,TM0} - dL_{eff,TM} \cdot n_{z,TE0}, \quad (7)$$

Such novel effect is exploited to realize the polarization-dependent phase shift. The 2×2 MMI can combine two beams with $\pi/2$ or $-\pi/2$ phase difference. For TE polarization, the phase shift

is chosen to be $d\varphi_{TE} = -\pi/2$ with the corresponding $dL_{eff,TE} = \lambda/4$, so the input TE_0 mode can be coupled into the O_1 port, as shown in Fig. 1(c). For TM polarization, the phase shift is chosen to be $d\varphi_{TM} = \pi/2$ with the corresponding $dL_{eff,TM} = -\lambda/4$, so the input TM_0 mode can be coupled into the O_2 port, as shown in Fig. 1(d). Thus, dL_1 and dL_2 can be determined by using the following equations:

$$dL_1 = \frac{\lambda \cdot (n_{y,TM0} + n_{y,TE0})}{4\Delta_0}, \quad (8)$$

$$dL_2 = \frac{\lambda \cdot (n_{z,TM0} + n_{z,TE0})}{4\Delta_0}, \quad (9)$$

Additionally, dL_1 and dL_2 should be positive values, so the constraint for Eqs. (8) and (9) can be written as:

$$\Delta_0 > 0, \quad (10)$$

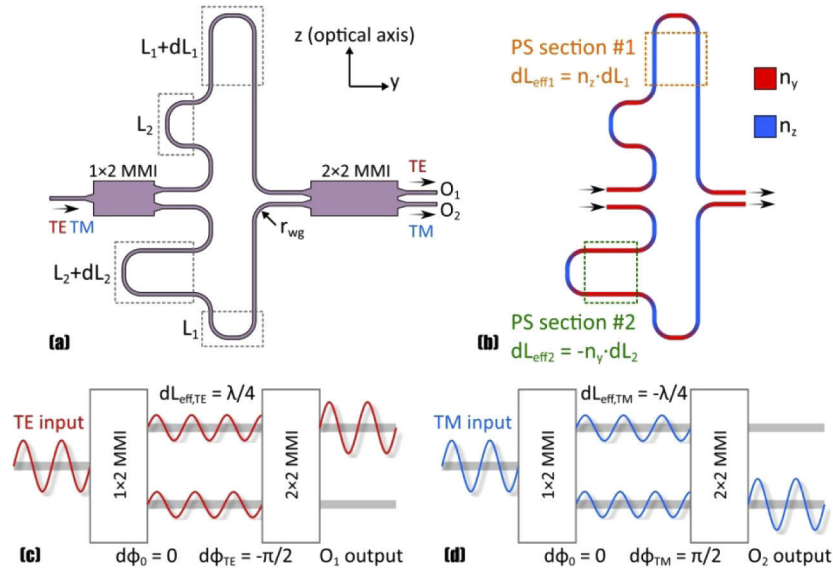


Fig. 1. (a) The schematic for the MZI-based PBS with some key parameters labeled. (b) The effective index distribution for the anisotropy-engineered phase shifter (PS). (c) (d) The schematics for the phase shift and light coupling in the MZI-based PBS.

We then calculate the effective indices for the TE_0 and TM_0 modes in the LNOI waveguide with varied waveguide width w_{wg} using the finite element method (FEM), as shown in Figs. 2(b) and 2(c). In this paper, the device is designed based on the LNOI with a LN top layer ($n_{LN,e} \approx 2.13$, $n_{LN,o} \approx 2.21$ at $1.55\mu\text{m}$ wavelength) and a $3\text{-}\mu\text{m}$ thick SiO_2 buffer layer ($n_{\text{SiO}_2} \approx 1.45$ at $1.55\mu\text{m}$ wavelength), as shown in Fig. 2(a). The material dispersions for LN and SiO_2 can be found in Refs. [33] and [34]. The cladding is chosen as a $2\text{-}\mu\text{m}$ thick SiO_2 layer. The thickness of the LN top layer is chosen to be $h_{LN} = 700\text{ nm}$ to ensure a strong light confinement and small bending radius. The relatively large h_{LN} could also make the PBS compatible with the LNOI-based modulator, since the thick LN top layer is also required to match the optical field and electric field in the high-speed modulation [9]. The side-wall tilt angle is set to be $\theta = 70^\circ$ considering the current state of fabrication technology. For TE polarization, the major electric component is parallel/perpendicular to the optical axis when propagating along y -/ z -direction, so $n_{y,TE0}$ is always lower than $n_{z,TE0}$, i.e., $n_{y,TE0} < n_{z,TE0}$, as shown in Fig. 2(b). For TM polarization, the major electric component is always perpendicular to the optical axis. However, the TM_0

mode also has an electric component along the propagation direction, so $n_{y, TM0}$ is always higher than $n_{z, TM0}$, i.e., $n_{y, TM0} > n_{z, TM0}$, as shown in Fig. 2(c). Thus, one can always find the $[dL_1, dL_2]$ combination to achieve the required phase shift since the constraint equation [Eq. (10)] is naturally satisfied for the LNOI waveguide. The waveguide width is then chosen to be $w_{wg} = 300$ nm to meet the single-mode requirement. One might find that the aspect ratio is relatively high for the optimized LNOI waveguide, which might increase the propagation loss due to the side-wall roughness induced by the dry etching. It should be noted that the propagation length is quite short for the PS section, so the total propagation loss is still negligible. Moreover, one can also use the bi-level mode converter to connect the PBS with the other LNOI devices based on the low-loss shallow-ridge waveguide [35]. Figure 2(d) shows the calculated TE_0 and TM_0 mode profiles for the optimized LNOI waveguide at $1.55 \mu\text{m}$ wavelength. We also calculate the effective indices for the TE_0 and TM_0 modes at different wavelengths, as shown in Figs. 2(e) and 2(f). A weak dispersion can be observed for both TE_0 and TM_0 modes over the 1.45 - $1.65 \mu\text{m}$ wavelength range. The PS length is then calculated to be $dL_1 = 13.64 \mu\text{m}$ and $dL_2 = 13.79 \mu\text{m}$ according to Eqs. (8) and (9). The 90° waveguide bends are utilized to connect the phase shifting sections with different directions [32,36,37]. The bending radius is optimized to be $r_{wg} = 25 \mu\text{m}$ to ensure the low bending loss $EL_{bend} < 0.01$ dB. Here, EL_{bend} denotes the excess loss when TE_0 mode propagates through a 90° waveguide bend. The reference arm length is chosen to be $L_1 = 90 \mu\text{m}$ and $L_2 = 90 \mu\text{m}$ to ensure a compact device size (bending lengths are also included).

The 1×2 MMI is then designed to achieve the power splitting, as shown in Fig. 3(a). For the 1×2 MMI, the input port (#1) is placed at the $z = 0$ position, while the output ports (#2 and #3) are placed at the $z = \pm w_{MMI1}/4$ positions. The adiabatic tapers are utilized to reduce the coupling loss in the MMI. The length and opening width of the adiabatic taper are chosen to be $L_{tp} = 2 \mu\text{m}$ and $w_{tp} = 700$ nm. Here, we choose a relatively small MMI width, so the multi-mode section can only support a few number of modes [38]. In such a few-mode MMI, the 50:50 splitting is mainly induced by the TE_0 - TE_2 or TM_0 - TM_2 interference. Thus, the MMI length can be obtained by using the following equations [39]:

$$L_{MMI1, TE} = \frac{\lambda}{4(n_{y, TE0} - n_{y, TE2})}, \quad (11)$$

$$L_{MMI1, TM} = \frac{\lambda}{4(n_{y, TM0} - n_{y, TM2})}, \quad (12)$$

where $L_{MMI1, TE}$ and $L_{MMI1, TM}$ are the MMI length for the TE and TM polarizations, $n_{y, TE0}/n_{y, TE2}/n_{y, TM0}/n_{y, TM2}$ are the effective indices for the $TE_0/TE_2/TM_0/TM_2$ modes propagating along y-direction. We calculate the MMI length L_{MMI1} with varied MMI width w_{MMI1} using Eqs. (11) and (12), as shown in Fig. 3(b). One can find that the difference between $L_{MMI1, TE}$ and $L_{MMI1, TM}$ is negligible, indicating the polarization-insensitive power splitting. The MMI width is chosen to be $w_{MMI1} = 4.5 \mu\text{m}$ to ensure a relatively large spacing between the output ports. The MMI length is then optimized to be $L_{MMI1} = (L_{MMI1, TE} + L_{MMI1, TM})/2 = 9.3 \mu\text{m}$. For the optimized 1×2 MMI, the excess losses are calculated to be $EL_{MMI1} < 0.5$ dB for both TE/TM polarizations over a 200-nm wavelength span, as shown in Fig. 3(c). Here, the excess loss EL_{MMI1} is defined as:

$$EL_{MMI1} = -10 \log_{10}(T_{\#2} + T_{\#3}), \quad (13)$$

where $T_{\#2}$ and $T_{\#3}$ are the transmittances at port #2 and #3 when the light is launched from port #1 [see Fig. 3(a)]. Figures 3(d) and 3(e) show the calculated light propagation profiles for the optimized 1×2 MMI. From the profiles, both TE_0 and TM_0 modes can be evenly split and coupled into the output ports with low losses.

We then optimize the 2×2 MMI to achieve the light coupling and polarization division, as shown in Fig. 4(a). For the 2×2 MMI, the input ports and output ports are placed at the

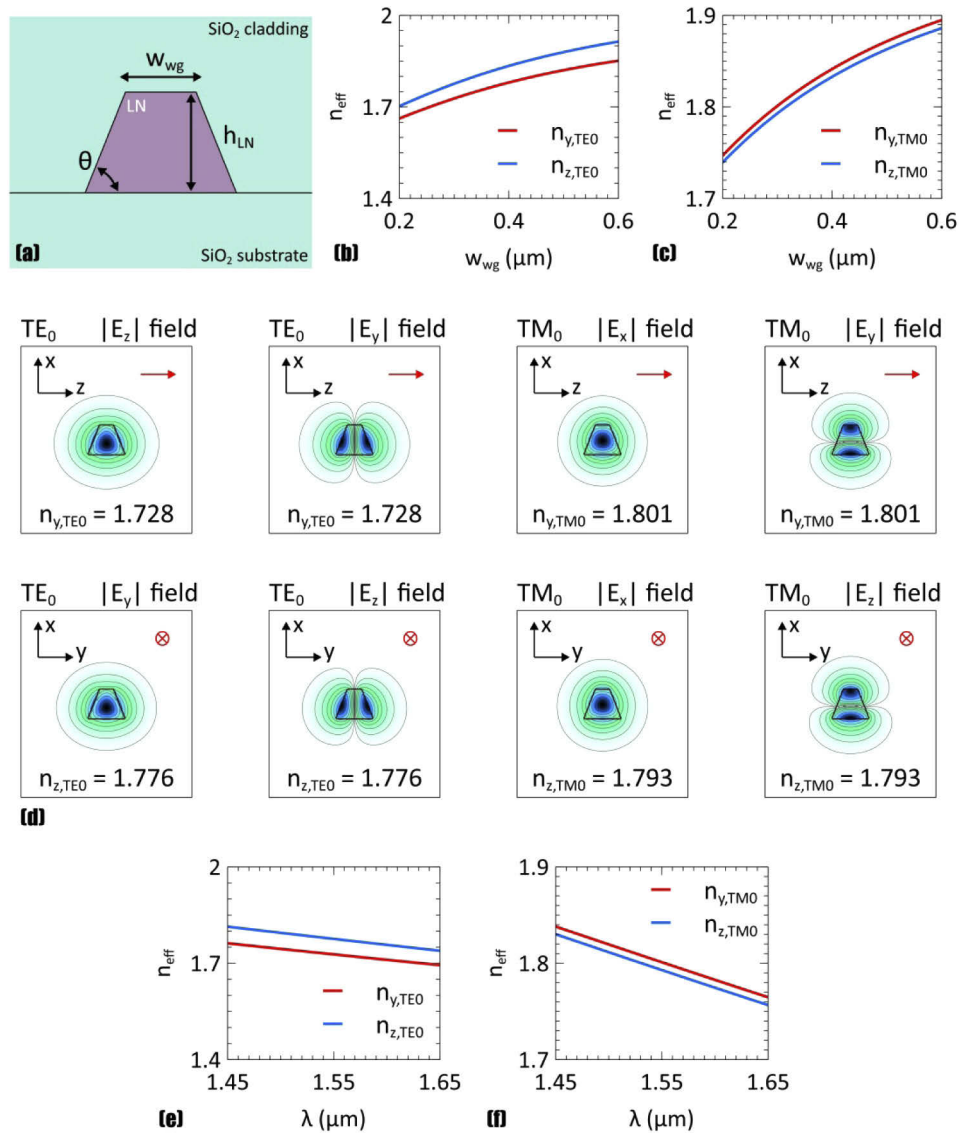


Fig. 2. (a) The cross-section of the single-mode LNOI waveguide with some key parameters labeled. The calculated (b) TE₀ and (c) TM₀ effective indices for the LNOI waveguide with varied waveguide width w_{wg} . (d) The calculated mode profiles for the TE₀ and TM₀ modes in the optimized LNOI waveguide with different directions. The red arrows show the optical axis orientations. The calculated dispersion curves for the (e) TE₀ and (f) TM₀ modes.

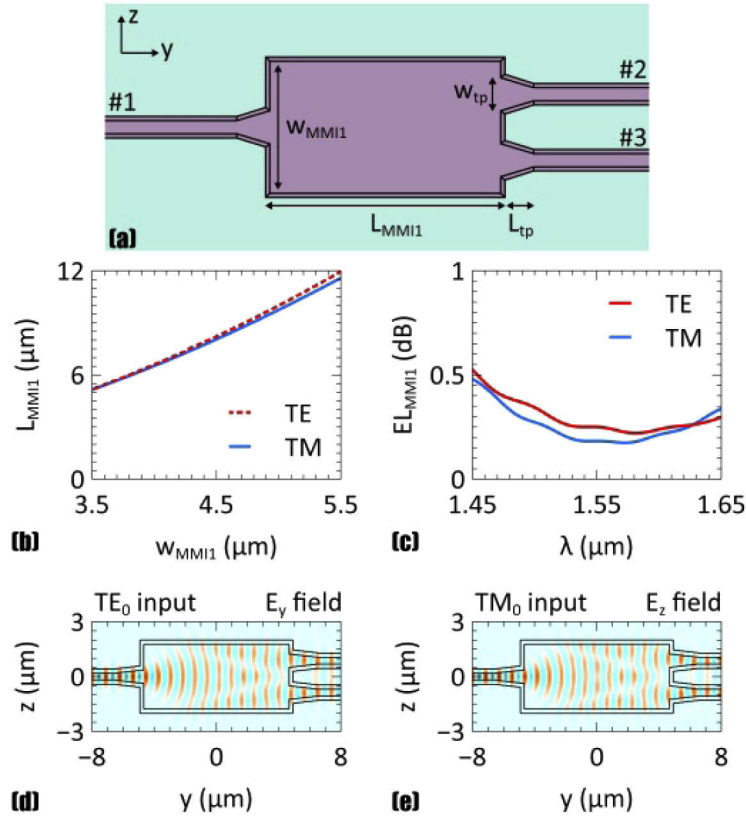


Fig. 3. (a) The schematic for the 1×2 MMI with some key parameters labeled. (b) The calculated MMI length L_{MMI1} with varied MMI width w_{MMI1} . (c) The calculated excess loss EL_{MMI1} for the optimized 1×2 MMI. The calculated light propagation profiles for the optimized 1×2 MMI when the (d) TE₀ and (e) TM₀ modes are launched.

$z = \pm w_{MMI2}/6$ positions. We calculate the MMI length L_{MMI2} with varied MMI width w_{MMI2} using the multi-mode interference theory [40,41]:

$$L_{MMI2,TE} = \frac{\lambda}{4(n_{y,TE0} - n_{y,TE1})}, \quad (14)$$

$$L_{MMI2,TM} = \frac{\lambda}{4(n_{y,TM0} - n_{y,TM1})}, \quad (15)$$

where $L_{MMI2,TE}$ and $L_{MMI2,TM}$ are the MMI length for the TE and TM polarizations, $n_{y,TE0}/n_{y,TE1}/n_{y,TM0}/n_{y,TM1}$ are the effective indices for the TE₀/TE₁/TM₀/TM₁ modes propagating along y-direction. The calculation results are shown in Fig. 4(b). It can be found that the calculated $L_{MMI2,TE}$ and $L_{MMI2,TM}$ are quite close, indicating the polarization-insensitive light coupling. The MMI width is chosen to be $w_{MMI2} = 4.5 \mu\text{m}$ to inhibit the crosstalk between output ports. The MMI length is then optimized to be $L_{MMI2} = (L_{MMI2,TE} + L_{MMI2,TM})/2 = 20.4 \mu\text{m}$. We then calculate the excess loss and coupling ratio for the optimized 2×2 MMI to evaluate the device performance. Here, the excess loss EL_{MMI2} and coupling ratio CR are defined as:

$$EL_{MMI2} = -10\log_{10}(T_{\#4} + T_{\#5}), \quad (16)$$

$$CR = \frac{T_{\#4}}{T_{\#4} + T_{\#5}}, \quad (17)$$

where $T_{\#4}$ and $T_{\#5}$ are the transmittances at ports #4 and #5 when light is launched from port #6 [see Fig. 4(a)]. From Fig. 4(c), the excess loss is calculated to be $EL_{MMI2} < 1$ dB for both TE/TM polarizations over the 1.50-1.65 μm wavelength range. Moreover, for the optimized 2 $\times$ 2 MMI, the coupling ratio is calculated to be $CR \approx 0.5$ over a 200-nm wavelength span, as shown in Fig. 4(d). Thus, the optimized 2 $\times$ 2 MMI should be able to combine the two beams launched from ports #4 and #5 since the device is reciprocal. To verify the above analysis, we calculate the light propagation profiles for the optimized 2 $\times$ 2 MMI, as shown in Figs. 4(e) and 4(f). For TE polarization with $d\phi_{TE} = -\pi/2$, the incident light can be completely coupled into port #6 [see Fig. 4(e)], while for TM polarization with $d\phi_{TM} = \pi/2$, the incident light can be completely coupled into port #7 [see Fig. 4(f)].

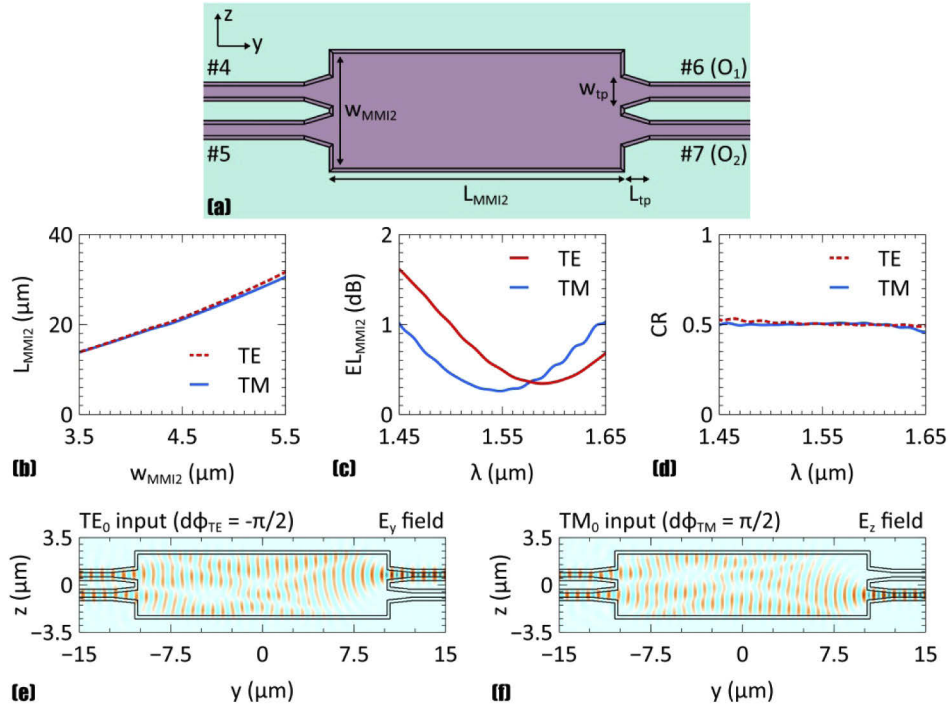


Fig. 4. (a) The schematic for the 2 $\times$ 2 MMI with some key parameters labeled. (b) The calculated MMI length L_{MMI2} with varied MMI width w_{MMI2} . The calculated (c) excess loss EL_{MMI2} and (d) coupling ratio CR for the optimized 2 $\times$ 2 MMI. The calculated light propagation profiles for the optimized 2 $\times$ 2 MMI when the (e) TE₀ and (f) TM₀ modes are launched.

The optimized parameters are summarized as following: $w_{wg} = 300$ nm, $L_1 = 90$ μm , $L_2 = 90$ μm , $dL_1 = 13.64$ μm , $dL_2 = 13.79$ μm , $w_{tp} = 700$ nm, $L_{tp} = 2$ μm , $w_{MMI1} = 4.5$ μm , $w_{MMI2} = 4.5$ μm , $L_{MMI1} = 9.3$ μm and $L_{MMI2} = 20.4$ μm . The transfer matrix method (TMM) is utilized to calculate the transmittance spectra for the optimized PBS, as shown in the 1st column of Fig. 5. It can be found that the incident TE₀ mode is coupled into the O₁ port, while the incident TM₀ mode is coupled into the O₂ port. At the central wavelength, the polarization extinction ratio is calculated to be $PER_{TE} = 47.7$ dB, $PER_{TM} = 48.0$ dB, while the calculated excess loss is as low as $EL_{TE} = 0.9$ dB and $EL_{TM} = 0.6$ dB. Here, the polarization extinction ratio PER and excess loss EL are defined as:

$$PER_{TE} = 10\log_{10}(T_{O1,TE}/T_{O2,TE}), \quad (18)$$

$$PER_{TM} = 10\log_{10}(T_{O2,TM}/T_{O1,TM}), \quad (19)$$

$$EL_{TE} = -10\log_{10}(T_{O1,TE}), \quad (20)$$

$$EL_{TM} = -10\log_{10}(T_{O2,TM}), \quad (21)$$

where T_{O1} and T_{O2} are the transmittances at the ports O_1 and O_2 . Moreover, the polarization extinction ratio is calculated to be $PER_{TE} > 17.8$ dB and $PER_{TM} > 22.8$ dB over an ultra-broad working bandwidth $BW > 200$ nm, which shows great compatibility with the wavelength-division multiplexing (WDM) systems [42–44]. One might find that some works have demonstrated for MZI-based PBSs on silicon previously [45,46]. However, the working principle for the silicon PBS is totally different from ours. Silicon is an isotropic material, i.e., $n_{y,TE0} = n_{z,TE0}$, $n_{y,TM0} = n_{z,TM0}$, so the phase shift cannot be arbitrarily controlled for TE and TM polarizations since Eq. (10) is not satisfied. Thus, one has to modify the waveguide width to realize the higher-order phase shift $d\varphi = (m \pm 1/2)\pi$. Thus, MZI-based silicon PBSs often suffer from a relatively low PER and a narrow bandwidth [45]. In contrast, in our proposed design, the zero-order phase shift $d\varphi = \pm\pi/2$ can be achieved by engineering the anisotropy of the LNOI waveguide, leading to an ultra-broad working bandwidth of $BW > 200$ nm. We also calculate the transmittance spectra with w_{wg} deviated from the optimized value, as shown in the 2nd and 3rd column of Fig. 5. It can be found that the high $PER > 15$ dB can still be maintained over the 200-nm bandwidth even with the large waveguide-width deviation of $dw_{wg} = \pm 30$ nm, indicating a large fabrication tolerance.

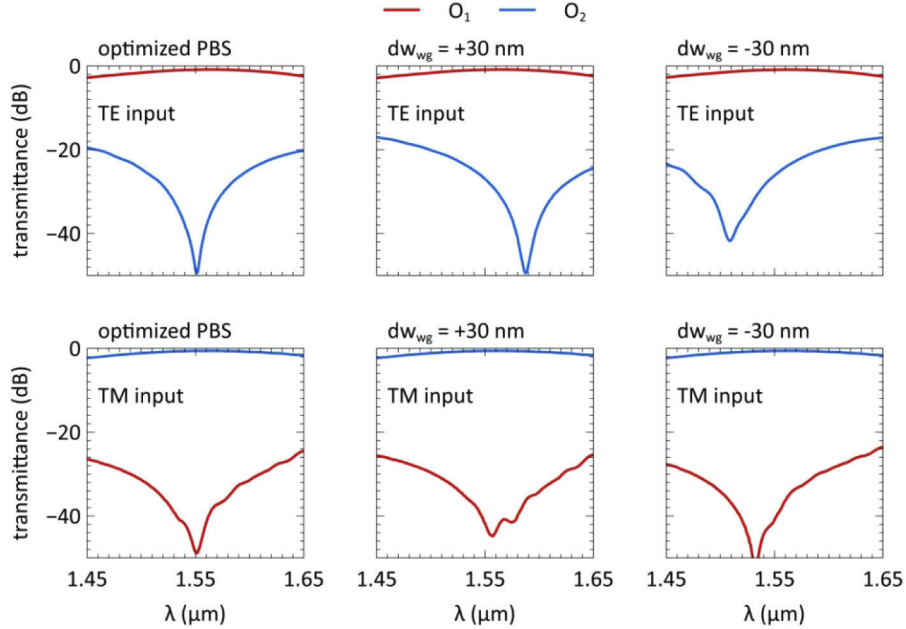


Fig. 5. The calculated transmittance spectra for the MZI-based PBS when the TE_0 and TM_0 modes are launched.

3. Conclusion

In conclusion, we have proposed an MZI-based PBS on the x-cut LNOI platform. The phase shift can be simultaneously controlled for the TE and TM polarizations by engineering the anisotropy of the LNOI waveguide. For TE polarization, the phase shift is $-\pi/2$, while for TM polarization, the phase shift is $\pi/2$, so the incident TE and TM modes can be coupled into different output ports. The calculation results show low $EL < 0.9$ dB, high $PER > 47.7$ dB and ultra-broad BW

> 200 nm. To the best of our knowledge, the proposed structure is the first integrated PBS on the x-cut LNOI platform, paving ways for the polarization-transparent photonic circuits and high-performance PDM transmitters.

Funding

National Natural Science Foundation of China (11861121002, 61922070, 61961146003); Major State Basic Research Development Program of China (2019YFB1803902).

Disclosures

The authors declare that there are no conflicts of interest related to this article.

References

1. G. Poberaj, H. Hu, W. Sohler, and P. Günter, "Lithium niobate on insulator (LNOI) for micro-photonic devices," *Laser & Photon. Rev.* **6**(4), 488–503 (2012).
2. I. Krasnokutskaya, J.-L. J. Tambasco, X. Li, and A. Peruzzo, "Ultra-low loss photonic circuits in lithium niobate on insulator," *Opt. Express* **26**(2), 897–904 (2018).
3. H. Hu, R. Ricken, and W. Sohler, "Lithium niobate photonic wires," *Opt. Express* **17**(26), 24261–24268 (2009).
4. S. Jin, L. Xu, H. Zhang, and Y. Li, "LiNbO₃ thin-film modulators using silicon nitride surface ridge waveguides," *IEEE Photonics Technol. Lett.* **28**(7), 736–739 (2016).
5. L. Cai, Y. Wang, and H. Hu, "Low-loss waveguides in a single-crystal lithium niobate thin film," *Opt. Lett.* **40**(13), 3013 (2015).
6. H. Liang, R. Luo, Y. He, H. Jiang, and Q. Lin, "High-quality lithium niobate photonic crystal nanocavities," *Optica* **4**(10), 1251 (2017).
7. R. Wu, J. Zhang, N. Yao, W. Fang, L. Qiao, Z. Chai, J. Lin, and Y. Cheng, "Lithium niobate micro-disk resonators of quality factors above 10^7 ," *Opt. Lett.* **43**(17), 4116–4119 (2018).
8. M. Zhang, C. Wang, R. Cheng, A. Shams-Ansari, and M. Lončar, "Monolithic ultra-high-Q lithium niobate microring resonator," *Optica* **4**(12), 1536 (2017).
9. C. Wang, M. Zhang, B. Stern, M. Lipson, and M. Lončar, "Nanophotonic lithium niobate electro-optic modulators," *Opt. Express* **26**(2), 1547–1555 (2018).
10. C. Wang, M. Zhang, X. Chen, M. Bertrand, A. Shams-Ansari, S. Chandrasekhar, P. Winzer, and M. Lončar, "Integrated lithium niobate electro-optic modulators operating at CMOS-compatible voltages," *Nature* **562**(7725), 101–104 (2018).
11. M. He, M. Xu, Y. Ren, J. Jian, Z. Ruan, Y. Xu, S. Gao, S. Sun, X. Wen, L. Zhou, L. Liu, C. Guo, H. Chen, S. Yu, L. Liu, and X. Cai, "High-performance hybrid silicon and lithium niobate Mach-Zehnder modulators for 100 Gbit s⁻¹ and beyond," *Nat. Photonics* **13**(5), 359–364 (2019).
12. A. Guarino, G. Poberaj, D. Rezzonico, R. Degl'Innocenti, and P. Günter, "Electro-optically tunable microring resonators in lithium niobate," *Nat. Photonics* **1**(7), 407–410 (2007).
13. A. J. Mercante, S. Shi, P. Yao, L. Xie, R. M. Weikle, and D. W. Prather, "Thin film lithium niobate electro-optic modulator with terahertz operating bandwidth," *Opt. Express* **26**(11), 14810–14816 (2018).
14. V. Stenger, J. Toney, A. Pollick, J. Busch, J. Scholl, P. Pontius, and S. Sriram, "Engineered Thin Film Lithium Niobate Substrate for High Gain-Bandwidth Electro-optic Modulators," in *CLEO: 2013, OSA Technical Digest (online) (Optical Society of America, 2013), paper CW3O.3*.
15. C. Wang, X. Xiong, N. Andrade, V. Venkataraman, X.-F. Ren, G.-C. Guo, and M. Lončar, "Second harmonic generation in nano-structured thin-film lithium niobate waveguides," *Opt. Express* **25**(6), 6963–6973 (2017).
16. J.-Y. Chen, Y. Sua, Z.-H. Ma, C. Tang, Z. Li, and Y.-P. Huang, "Efficient parametric frequency conversion in lithium niobate nanophotonic chips," *OSA Continuum* **2**(10), 2914 (2019).
17. D. Pohl, M. Escalé, M. Madi, F. Kaufmann, P. Brotzer, A. Sergeev, B. Guldemann, P. Giaccari, E. Alberti, U. Meier, and R. Grange, "An integrated broadband spectrometer on thin-film lithium niobate," *Nat. Photonics* **14**(1), 24–29 (2020).
18. H. Fukuda, K. Yamada, T. Tsuchizawa, T. Watanabe, H. Shinjima, and S.-I. Itabashi, "Silicon photonic circuit with polarization diversity," *Opt. Express* **16**(7), 4872 (2008).
19. J. Wang, S. He, and D. Dai, "On-chip silicon 8-channel hybrid (de)multiplexer enabling simultaneous mode- and polarization-division-multiplexing," *Laser Photon. Rev.* **8** (2014).
20. L.-T. Feng, M. Zhang, Z.-Y. Zhou, M. Li, X. Xiong, L. Yu, B.-S. Shi, G.-P. Guo, D.-X. Dai, X.-F. Ren, and G.-C. Guo, "On-chip coherent conversion of photonic quantum entanglement between different degrees of freedom," *Nat. Commun.* **7**(1), 11985 (2016).
21. D. Dai, Z. Wang, and J. E. Bowers, "Ultrashort broadband polarization beam splitter based on an asymmetrical directional coupler," *Opt. Lett.* **36**(13), 2590 (2011).
22. H. Wu, Y. Tan, and D. Dai, "Ultra-broadband high-performance polarizing beam splitter on silicon," *Opt. Express* **25**(6), 6069 (2017).

23. H. Xu and Y. Shi, "Ultra-broadband silicon polarization splitter-rotator based on the multi-mode waveguide," *Opt. Express* **25**(15), 18485–18491 (2017).
24. H. Xu and Y. Shi, "On-chip silicon TE-pass polarizer based on asymmetrical directional couplers," *IEEE Photonics Technol. Lett.* **29**(11), 861–864 (2017).
25. Y. Zhang, Y. He, J. Wu, X. Jiang, R. Liu, C. Qiu, X. Jiang, J. Yang, C. Tremblay, and Y. Su, "High-extinction-ratio silicon polarization beam splitter with tolerance to waveguide width and coupling length variations," *Opt. Express* **24**(6), 6586–6593 (2016).
26. X. Guan, H. Wu, Y. Shi, and D. Dai, "Extremely small polarization beam splitter based on a multimode interference coupler with a silicon hybrid plasmonic waveguide," *Opt. Lett.* **39**(2), 259–262 (2014).
27. Y. Jiao, D. Dai, Y. Shi, and S. He, "Shortened polarization beam splitters with two cascaded multimode interference sections," *IEEE Photonics Technol. Lett.* **21**(20), 1538–1540 (2009).
28. C. Li and D. Dai, "Compact polarization beam splitter for silicon photonic integrated circuits with a 340-nm-thick silicon core layer," *Opt. Lett.* **42**(21), 4243 (2017).
29. H. Xu, D. Dai, and Y. Shi, "Ultra-broadband and ultra-compact on-chip silicon polarization beam splitter by using hetero-anisotropic metamaterials," *Laser Photon. Rev.* **13**(4), 1800349 (2019).
30. H. Xu, D. Dai, and Y. Shi, "Anisotropic metamaterial-assisted all-silicon polarizer with 415-nm bandwidth," *Photonics Res.* **7**(12), 1432 (2019).
31. H.-P. Chung, C.-H. Lee, K.-H. Huang, S.-L. Yang, K. Wang, A. S. Solntsev, A. A. Sukhorukov, F. Setzpfandt, and Y.-H. Chen, "Broadband on-chip polarization mode splitters in lithium niobate integrated adiabatic couplers," *Opt. Express* **27**(2), 1632 (2019).
32. A. Pan, C. Hu, C. Zeng, and J. Xia, "Fundamental mode hybridization in a thin film lithium niobate ridge waveguide," *Opt. Express* **27**(24), 35659–35669 (2019).
33. D. S. Smith, H. D. Riccius, and R. P. Edwin, "Refractive indices of lithium niobate," *Opt. Commun.* **17**(3), 332–335 (1976).
34. I. H. Malitson, "Interspecimen comparison of the refractive index of fused silica," *J. Opt. Soc. Am.* **55**(10), 1205–1209 (1965).
35. L. He, M. Zhang, A. Shams-Ansari, R. Zhu, C. Wang, and L. Marko, "Low-loss fiber-to-chip interface for lithium niobate photonic integrated circuits," *Opt. Lett.* **44**, 2314–2317 (2019).
36. X. Li, K. Chen, and Z. Hu, "Low-loss bent channel waveguides in lithium niobate thin film by proton exchange and dry etching," *Opt. Mater. Express* **8**(5), 1322 (2018).
37. M. Bahadori, Y. Yang, L. L. Goddard, and S. Gong, "High performance fully etched isotropic microring resonators in thin-film lithium niobate on insulator platform," *Opt. Express* **27**(15), 22025–22039 (2019).
38. R. Halir, G. Roelkens, A. Ortega-Monux, and J. G. Wangüemert-Pérez, "High-performance 90° hybrid based on a silicon-on-insulator multimode interference coupler," *Opt. Lett.* **36**(2), 178–180 (2011).
39. L. B. Soldano and E. C. M. Pennings, "Optical multi-mode interference devices based on self-imaging: principles and applications," *J. Lightwave Technol.* **13**(4), 615–627 (1995).
40. J. M. Heaton and R. M. Jenkins, "General matrix theory of self-imaging in multimode interference (MMI) couplers," *IEEE Photonics Technol. Lett.* **11**(2), 212–214 (1999).
41. D. J. Thomson, Y. Hu, G. T. Reed, and J.-M. Fedeli, "Low loss MMI couplers for high performance MZI modulators," *IEEE Photonics Technol. Lett.* **22**(20), 1485–1487 (2010).
42. H. Xu, L. Liu, and Y. Shi, "Polarization-insensitive four-channel coarse wavelength-division (de)multiplexer based on Mach-Zehnder interferometers with bent directional couplers and polarization rotators," *Opt. Lett.* **43**(7), 1483–1486 (2018).
43. H. Xu and Y. Shi, "Flat-top CWDM (de)multiplexer based on MZI with bent directional couplers," *IEEE Photonics Technol. Lett.* **30**(2), 169–172 (2018).
44. H. Xu and Y. Shi, "On-chip silicon triplexer based on asymmetrical directional couplers," *IEEE Photonics Technol. Lett.* **29**(15), 1265–1268 (2017).
45. D. Dai, Z. Wang, J. Peters, and J. E. Bowers, "Compact polarization beam splitter using an asymmetrical Mach-Zehnder interferometer based on silicon-on-insulator waveguides," *IEEE Photonics Technol. Lett.* **24**(8), 673–675 (2012).
46. D. Dai, Z. Wang, and J. E. Bowers, "Considerations for the design of asymmetrical Mach-Zehnder interferometers used as polarization beam splitters on a submicrometer silicon-on-insulator platform," *J. Lightwave Technol.* **29**(12), 1808–1817 (2011).